Clean CPU evidence: 0.6.0-clean-041f2a1

Measured source `041f2a158c3e51340b5b9e6df6886fa52584f9b5`, tree `4de2d0afb43c2127b26a5ed51d77d089016271d2`; dirty before/after: false. Library source: workspace 041f2a158c3e51340b5b9e6df6886fa52584f9b5. One pinned CPU; f64 operations and explicitly named f32 constructors; row-major inputs.

Workload / library	Per-call p50 / p95 (us)	Batch-mean p50 / p95 (us)	Rust allocations	Peak additional requested bytes	Process peak RSS (MiB)
cholesky_factor_64_f64/faer	7.904 / 8.040	8.188 / 8.705	2	33280	4.16
cholesky_factor_64_f64/kernix	22.931 / 25.260	24.261 / 30.336	2	65536	4.12
cholesky_factor_64_f64/nalgebra	13.031 / 14.325	13.621 / 14.861	1	32768	3.91
cholesky_factor_64_f64/numpy	10.924 / 17.209	10.749 / 11.189	unavailable	unavailable	32.58
create_ones_64x64_f32/candle	0.238 / 0.250	0.226 / 0.232	7	16696	4.59
create_ones_64x64_f32/faer	0.171 / 0.181	0.169 / 0.180	1	16384	3.86
create_ones_64x64_f32/kernix	0.183 / 0.277	0.188 / 0.194	1	16384	3.87
create_ones_64x64_f32/ndarray	0.155 / 0.156	0.145 / 0.150	1	16384	3.81
create_ones_64x64_f32/numpy	1.221 / 1.427	1.300 / 1.448	unavailable	unavailable	32.02
create_zeros_64x64_f32/candle	0.152 / 0.153	0.157 / 0.160	5	16648	4.34
create_zeros_64x64_f32/faer	0.190 / 0.194	0.166 / 0.199	1	16384	4.03
create_zeros_64x64_f32/kernix	0.134 / 0.137	0.140 / 0.144	1	16384	3.74
create_zeros_64x64_f32/ndarray	0.099 / 0.100	0.094 / 0.100	1	16384	3.70
create_zeros_64x64_f32/numpy	0.351 / 0.379	0.387 / 0.395	unavailable	unavailable	32.03
matmul_128_f64/candle	69.019 / 82.916	71.439 / 75.780	9	131352	5.36
matmul_128_f64/faer	76.137 / 78.465	77.306 / 78.972	1	131072	4.64
matmul_128_f64/kernix	113.767 / 127.927	116.137 / 122.255	3	327680	4.45
matmul_128_f64/nalgebra	110.415 / 122.328	112.113 / 118.237	2	327680	4.07
matmul_128_f64/ndarray	117.314 / 125.183	115.904 / 131.648	2	327680	4.37
matmul_128_f64/numpy	72.997 / 79.055	75.236 / 79.746	unavailable	unavailable	33.68
matmul_32_f64/candle	1.505 / 1.696	1.500 / 1.534	9	8592	4.92
matmul_32_f64/faer	1.493 / 1.546	1.480 / 1.721	1	8192	4.16
matmul_32_f64/kernix	1.541 / 1.944	1.563 / 1.644	3	24576	3.98
matmul_32_f64/nalgebra	1.494 / 1.598	1.428 / 1.569	2	24576	3.73
matmul_32_f64/ndarray	1.407 / 1.555	1.451 / 1.666	2	24576	3.64
matmul_32_f64/numpy	2.426 / 3.284	2.483 / 2.569	unavailable	unavailable	32.36
matmul_sum_backward_32x16x8_f64/candle	3.914 / 4.704	4.026 / 4.271	113	27988	5.59
matmul_sum_backward_32x16x8_f64/kernix	3.165 / 3.745	3.020 / 3.206	19	18376	4.00
qr_256x64_f64/faer	393.041 / 426.261	404.782 / 418.253	7	331776	5.08
qr_256x64_f64/kernix	468.986 / 524.363	480.923 / 519.950	6	557568	4.88
qr_256x64_f64/nalgebra	356.914 / 382.462	358.303 / 409.421	4	295424	4.27
qr_256x64_f64/numpy	326.424 / 351.262	329.967 / 343.904	unavailable	unavailable	33.51
qr_96x32_f64/faer	69.906 / 95.294	66.129 / 69.041	7	67584	4.54
qr_96x32_f64/kernix	45.810 / 49.252	46.038 / 46.735	6	106752	4.06
qr_96x32_f64/nalgebra	37.619 / 41.175	38.282 / 55.679	4	57600	3.66
qr_96x32_f64/numpy	36.491 / 39.224	38.338 / 39.856	unavailable	unavailable	32.70
softmax_256x64_f64/candle	77.876 / 83.362	75.333 / 92.035	37	265200	5.69
softmax_256x64_f64/kernix	84.758 / 97.453	85.652 / 90.543	1	131072	4.30

Workload / library	Per-call p50 / p95 (us)	Batch-mean p50 / p95 (us)	Rust allocations	Peak additional requested bytes	Process peak RSS (MiB)
softmax_256x64_f64/ndarray	67.896 / 72.941	69.009 / 71.483	1	131072	4.25
softmax_256x64_f64/numpy	97.789 / 118.113	97.894 / 108.880	unavailable	unavailable	32.61

Method and limits

- Per-call quantiles use nearest ranks over 256 individual warmed calls; batch quantiles use 15 means over separate 10-ms batches. They are separate distributions. Raw samples remain in raw.json.
- Eight warmup calls follow validation. Timing includes allocation, dispatch, computation and output destruction. The first call field excludes output destruction. Inputs and library/runtime imports are prepared outside timing.
- Three extra processes record their first operation calls. These are first-use observations, not cache-flushed or device-startup latency. Timer-pair overhead is reported without subtraction; tiny calls are clock-sensitive.
- ndarray and Candle softmax are explicit stable compositions, not claims about a native softmax API.
- Constructor cases allocate owned 64x64 f32 tensors. Pre-0.6 registry baselines use full(value), explicitly marked in metadata; current workspace runs use zeros/ones. No nested literal reinterpretation is involved.
- All GEMM inputs are C-order and match elementwise. Native output layout differs by library. QR materializes thin Q and R in every implementation. Cholesky returns reusable factors (or NumPy L); no extra factor materialization is timed.
- Kernix checks finite values/symmetry. Some competitors validate a narrower contract. No finite-data precondition check was removed to improve timing. Accuracy results and budgets are retained per worker.
- The backward case uses an actual reverse-mode graph in Kernix and Candle for sum(A @ W), with A constant and W trainable. Both copy input/weight data and returned gradients; no manual derivative is used as a framework ranking.
- Allocations are measured in a separately compiled instrumented binary after warmup, outside all timing loops. Counts cover Rust global-allocator requests, not native allocator internals, stack memory or all NumPy allocations. NumPy allocation counts are unavailable, not zero.
- Requested-byte peak is additional live Rust allocation during one call, including output; it is not scratch-only memory or peak RSS. RSS is the process high-water mark including fixtures, validation and runtime, read from Linux VmHWM. Other platforms may report unavailable.
- The published cases are allocation-inclusive CPU APIs and graph workloads. Pure internal kernel-only and end-to-end application/model latency are not measured. No workload result implies a complete application frame rate.
- faer is explicitly serial; Rayon and BLAS thread counts are one. No GPU comparison. Python/NumPy timings include Python dispatch. Backend and dependency versions/configuration, governor and RAM are recorded.
- A shared laptop has uncontrolled load/frequency/cache effects. Report comparator wins and losses; no universal ranking follows. The source archive comes from committed bytes, with result directories excluded. Library dependencies are pinned by the saved driver lockfile.